

32-bit RX Microcontrollers

The Core Difference



BY: BERND WESTHOFF

Manufacturers of industrial and consumer applications face pressure to drive down costs. 8- and 16-bit microcontroller units (MCUs) have become the mainstay of many embedded systems for their cost-effectiveness. More and more developers are leveraging on the platform concept to do so. Many industrial applications have been designed for platform development where costs can be reduced by reusing elements that have already been developed.

Renesas Electronics, a world leader in microcontrollers, has launched the RX MCU platform to provide 16-/32-bit performance at 8-/16-bit cost.

Here is how it achieves this.

Improved code efficiency

Lengthy codes require more on-chip memory to store. This, in turn, drives up system costs. The RX achieves code efficiency, and therefore lower system cost, using shorter instruction codes. The RX architecture has a set of 89 variable byte-length instructions. Frequently used instructions are identified, enhanced and adopted for the RX CPU.

Most of the frequently used instructions have been successfully limited to between one and four bytes. System engineers can now write application programs using fewer instructions than in the past, thanks to other aspects of the RX architecture.

Among those features are general-purpose registers, a three-operand format and extra addressing modes, including indexed register-indirect addressing. Benchmark tests have shown that RX devices gain a 35 per cent improvement in code efficiency in several different sample application solution sets, like data-communications and industrial equipment.

200MHz maximum operating frequency

The RX CPU core is fundamentally a new design, right down to the hardware level, including the bus configuration. The maximum operating frequency has been increased to 200 MHz. Its Harvard-type architecture allows instruction-fetch and memory-access operations to be performed independently. Pipeline performance is also improved through concurrent execution in a five-stage pipeline. Also, the CPU structure is designed for high-speed processing.

Support is provided for out-of-order completion by ensuring that no pipeline stall occurs in subsequent instructions, provided they have no influence on the following instructions. There are sixteen 32-bit general-purpose registers, nine 32-bit control registers and a 64-bit accumulator. The hardware that executes the core processing operations is improved, as well. Two examples of enhancements are the inclusion of an FPU (floating-point unit) and DSP functions.

Multi-bus configuration

A multi-bus approach for the RX bus architecture permits concurrent operation by providing two special-purpose internal buses—internal main bus 1 and internal main bus 2—and by separating the CPU and other bus masters. The new exDMAC (on-chip external direct memory access controller) added to RX chips allows data transfers via the external bus to operate independently. This feature allows the RX device to drive a QVGA or WQVGA display (480×272 pixel) without additional display driver IC and at a low workload level for the CPU.

High-speed MONOS flash memory

Accessing the Flash without wait state insertion is one of the most fundamental features of RX MCUs. Renesas engineers decided to take on this challenge because the CPU speeds of today's standard microcomputers have been increasing from 50 MHz to 100 MHz. Normally, the access speed for the embedded flash memory is not more than about 30 MHz. In such situations, the slow on-chip flash memory reduces the CPU's effective

Key enhancements in RX microcomputers

1. Improved code efficiency
2. 200MHz maximum operating frequency
3. Multi-bus configuration
4. High-speed MONOS flash memory
5. 90nm fabrication process with low power consumption
6. Extreme connectivity
7. Long lifetime by design
8. RX platform product line-up